# 9.4 Temporal Segment Networks (TSN)

- an obvious problem of the two-stream CNNs in their previous forms is their inability in modeling long-range temporal structure
- this is mainly due to their limited access to temporal context, as they are designed to operate only on a single frame (spatial networks) or a single stack of frames in a short snippet (temporal network)
- however, complex actions, such as sports actions, comprise multiple stages spanning over a relatively long time
- it would be quite a loss failing to utilize long-range temporal structures in these actions into CNN training
- to tackle this issue, temporal segment network is proposed, a video-level framework as shown in Figure 9, to enable to model dynamics throughout the whole video

# 9.4 Temporal Segment Networks (TSN)

- specifically, the proposed temporal segment network framework, aiming to utilize the visual information of entire videos to perform video-level prediction, is also composed of spatial stream CNNs and temporal stream CNNs
- instead of working on single frames or frame stacks, temporal segment networks operate on a sequence of short snippets, sparsely sampled from the entire video
- each snippet in this sequence will produce its own preliminary prediction of the action classes
- then, a consensus among the snippets will be derived as the video-level prediction
- in the learning process, the loss values of video-level predictions, other than those of snippet-level predictions, which were used in two-stream CNNs, are optimized by iteratively updating the model parameters

# 9.4 Temporal Segment Networks (TSN)

- formally, given a video $V$, we divide it into $K$ segments $\{S_1, S_2, \ldots, S_K\}$ of equal durations
- then, the temporal segment network models a sequence of snippets as follows:

$$TSN(T_1, T_2, \ldots, T_K) = \mathcal{H}(\mathcal{G}(\mathcal{F}(T_1; \boldsymbol{W}), F(T_2; \boldsymbol{W}), \ldots, F(T_K; \boldsymbol{W}))).$$

- here $(T_1, T_2, \ldots, T_K)$ is a sequence of snippets
- each snippet $T_k$ is randomly sampled from its corresponding segment $S_k$
- $\mathcal{F}(T_k; \boldsymbol{W})$ is the function representing a CNN with parameters $\boldsymbol{W}$, which operates on the short snippet $T_k$ and produces class scores for all the classes
- the segmental consensus function $\mathcal{G}$ combines the outputs from multiple short snippets to obtain a consensus of class hypothesis among them
- based on this consensus, the prediction function $\mathcal{H}$ predicts the probability of each action class for the whole video
- here, the widely used softmax function for $\mathcal{H}$ is chosen

# 9.4 Temporal Segment Networks (TSN)

- combining with standard categorical cross-entropy loss, the final loss function regarding the segmental consensus $\boldsymbol{G} = \mathcal{G}(\mathcal{F}(T_1; \boldsymbol{W}), F(T_2; \boldsymbol{W}), \ldots, F(T_K; \boldsymbol{W}))$ is formed as

$$\mathcal{L}(y, \boldsymbol{G}) = -\sum_{i=1}^{C} y_i \left( G_i - \log \left( \sum_{j=1}^{C} \exp G_j \right) \right),$$

  where $C$ is the number of action classes and $y_i$ the ground truth label concerning class $i$
- the number of snippets $K$ is set to 3
- the form of consensus function $\mathcal{G}$ remains an open question
- in the paper, the simplest form of $\mathcal{G}$ is used, where $G_i = g(\mathcal{F}_i(T_1), \ldots, \mathcal{F}_i(T_K))$
- here, a class score $G_i$ is inferred from the scores of the same class on all the snippets, using an aggregation function $g$
- several different forms of the aggregation function $g$, including evenly averaging, maximum, and weighted averaging were empirically evaluated
- among them, evenly averaging had the best performance

# 9.4 Temporal Segment Networks (TSN)

- this temporal segment network is differentiable or at least has subgradients, depending on the choice of $g$
- this allows utilizing the multiple snippets to jointly optimize the model parameters $\boldsymbol{W}$ with standard backpropagation algorithms
- in the backpropagation process, the gradients of model parameters $\boldsymbol{W}$ with respect to the loss value $\mathcal{L}$ can be derived as

$$\frac{\partial \mathcal{L}(y, \boldsymbol{G})}{\partial \boldsymbol{W}} = \frac{\partial \mathcal{L}}{\partial \boldsymbol{G}} \sum_{k=1}^{K} \frac{\partial \mathcal{G}}{\partial \mathcal{F}(T_k)} \frac{\partial \mathcal{F}(T_k)}{\partial \boldsymbol{W}}, \tag{3}$$

where $K$ is the number of segments the temporal segment network uses

# 9.4 Temporal Segment Networks (TSN)

- when a gradient-based optimization method, like stochastic gradient descent (SGD), is used to learn the model parameters, relation (3) guarantees that the parameter updates are utilizing the segmental consensus $\boldsymbol{G}$ derived from all snippet-level prediction
- optimized in this manner, the temporal segment network can learn model parameters from the entire video, rather than a short snippet
- meanwhile, by fixing $K$ for all videos, a sparse temporal sampling strategy is assembled, where the sampled snippets contain only a small portion of the frames
- this drastically reduces the computational cost for evaluating CNNs on the frames, compared with previous works using densely sampled frames

# 9.4 Temporal Segment Networks (TSN)

- temporal segment network provides a solid framework to perform video-level learning, but, to achieve optimal performance, a few practical concerns have to be taken care of, for example the limited number of training samples
- to this end, we study a series of good practices in training deep CNNs on video data, which are also directly applicable in learning temporal segment networks

# 9.4 Temporal Segment Networks (TSN)

- **Network architecture** is an important factor in neural network design
- several works have shown that deeper structures improve object recognition performance
- however, the original two-stream CNNs employed a relatively shallow network structure
- in the paper, the Inception with Batch Normalization (BN-Inception) is chosen as building block, due to its good balance between accuracy and efficiency
- the original BN-Inception architecture is adapted to the design of two-stream CNNs
- like in the original two-stream CNNs, the spatial stream CNN operates on a single RGB image, and the temporal stream CNN takes a stack of consecutive optical flow fields as input

# 9.4 Temporal Segment Networks (TSN)

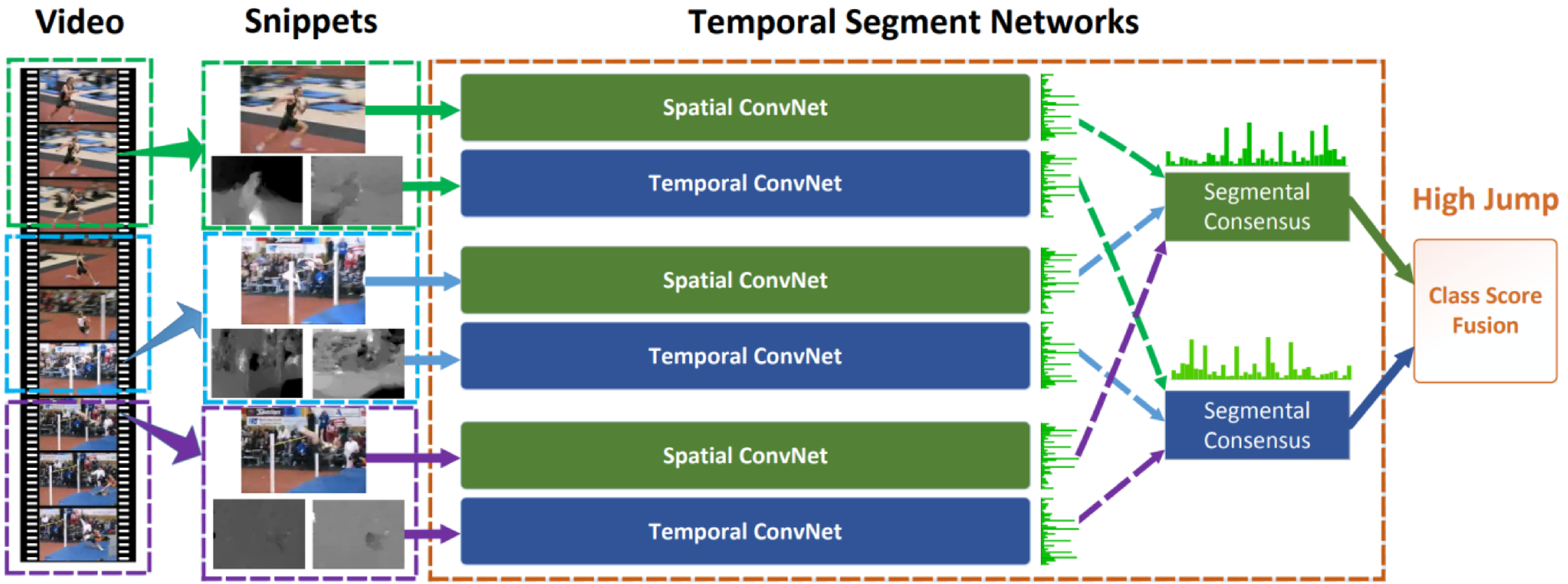



Figure 9: Temporal segment network: one input video is divided into $K$ segments and a short snippet is randomly selected from each segment. The class scores of different snippets are fused by the segmental consensus function to yield segmental consensus, which is a video-level prediction. Predictions from all modalities are then fused to produce the final prediction. CNNs on all snippets share parameters.

# 9.4 Temporal Segment Networks (TSN)

- **Network inputs**: we are also interested in exploring more input modalities to enhance the discriminative power of temporal segment networks
- originally, the two-stream CNNs used RGB images for the spatial stream and stacked optical flow fields for the temporal stream
- here, two extra modalities are proposed, namely *RGB difference* and *warped optical flow fields*

# 9.4 Temporal Segment Networks (TSN)

- a single RGB image usually encodes static appearance at a specific time point and lacks the contextual information about previous and next frames
- as shown in Figure 10, RGB difference between two consecutive frames describes the appearance change, which may correspond to the motion salient region
- thus, adding stacked RGB difference as another input modality is experimented with, and its performance in action recognition is investigated

# 9.4 Temporal Segment Networks (TSN)

- the temporal stream CNNs take optical flow field as input and aim to capture the motion information
- in realistic videos, however, there usually exists camera motion, and optical flow fields may not concentrate on the human action
- as shown in Figure 10, a remarkable amount of horizontal movement is highlighted in the background due to the camera motion
- warped optical flow fields are proposed as an additional input modality
- the warped optical flow is extracted by first estimating the homography matrix and then compensating camera motion
- as shown in Figure 10, the warped optical flow suppresses the background motion and makes motion concentrate on the actor

# 9.4 Temporal Segment Networks (TSN)

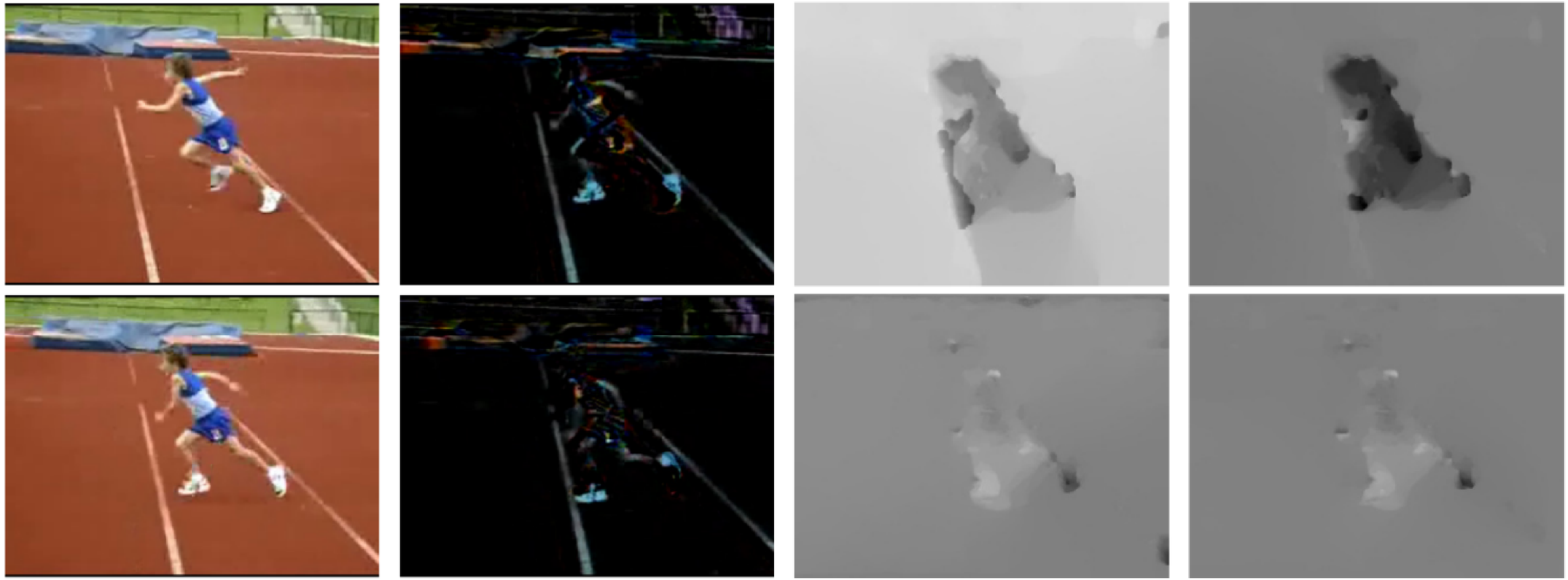


Figure 10: Examples of four types of input modality: RGB images, RGB difference, optical flow fields ($x$, $y$ directions), and warped optical flow fields ($x$, $y$ directions).